Computational Statistics and Simulation

Worksheet 4: Maximum Likelihood Estimation

Academic year 2025/26

1. Consider the Pareto model with shape parameter α and scale parameter β, defined by the distribution function

$$F(x) = 1 - \left(\frac{\beta}{x}\right)^{\alpha}, \qquad x \geq \beta \quad (\alpha > 0,\ \beta > 0).$$

   (a) Take $\beta = 1$. Using the R functions `mle`, `optimize` and `uniroot` from the `stats4` package, obtain maximum likelihood estimates of α, generating a sample of size $n = 1000$ and assuming the model with shape parameter $\alpha = 2$. Compare with the value given by the expression of the estimator.

   (b) Now suppose both parameters are unknown:

      i. using the R functions and generating a sample of size $n = 1000$ from the $Pa(2, 5)$ model, obtain maximum likelihood estimates of the parameters;

      ii. solve the previous problem applying it to the data set:

$$1.2,\ 1.8,\ 2.5,\ 3.2,\ 4.5,\ 5.9,\ 13.5,\ 22.3,\ 96.1.$$

2. Let $x = (x_1, \ldots, x_n)$ be an observed sample from the $Beta(a, b)$ population, defined by the p.d.f.

$$f(x; \theta) = f(x; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}, \qquad 0 < x < 1, \quad (a, b > 0).$$

   Determine

   (a) expressions for the maximum likelihood estimators of the parameters;

   (b) maximum likelihood estimates, using the `optim` function, from a sample generated from the $Beta(3, 2)$.

3. Consider the problem of estimating the parameters of the quadratic form of centred Gaussian random variables given by

$$Y = \lambda_1 X_1^2 + \cdots + \lambda_k X_k^2,$$

   where $X_j \sim N(0, 1)$, $j = 1, \ldots, k$ and $\lambda_1 > \cdots > \lambda_k > 0$.

   Since $Y_j = \lambda_j X_j^2 \sim Ga\left(\frac{1}{2}, \frac{1}{2\lambda_j}\right)$, $j = 1, \ldots, k$, we have

$$Y = \frac{1}{k} Ga\left(\frac{1}{2}, \frac{1}{2\lambda_1}\right) + \cdots + \frac{1}{k} Ga\left(\frac{1}{2}, \frac{1}{2\lambda_k}\right).$$

   Assume $\sum \lambda_j = 1$, $k = 3$, and a random sample $(y_1, \ldots, y_{200})$ from the model with distribution Y and parameters $\lambda_1 = 0.6$, $\lambda_2 = 0.25$, $\lambda_3 = 0.15$.

   Obtain maximum likelihood estimates of λ_j, $j = 1, 2, 3$.